

```

    'Department': ['HR', 'IT', 'IT', 'Marketing', 'HR']
}
df = pd.DataFrame(data)
print("--- Numeric Summary ---")
print(df.describe())
print("\n--- Categorical Summary ---")
print(df.describe(include='object'))

```

```

--- Numeric Summary ---
      Age      Salary
count  5.000000  5.000000
mean   35.000000  77000.000000
std     7.905694  23874.672773
min    25.000000  50000.000000
25%    30.000000  60000.000000
50%    35.000000  75000.000000
75%    40.000000  90000.000000
max    45.000000  110000.000000

--- Categorical Summary ---
      Department
count          5
unique         3
top           HR
freq           2

```

Start coding or [generate](#) with AI.

```

import pandas as pd
data = {
    'Name': ['Alice', 'Bob', 'Charlie', 'David', 'Eva', 'Frank', 'Grace'],
    'Score': [85, 92, 78, 95, 88, 72, 91]
}
df = pd.DataFrame(data)
print(df.head())
print(df.head(2))
print(df.tail(3))

```

```

      Name  Score
0  Alice    85
1   Bob    92
2 Charlie    78
3  David    95
4   Eva    88
      Name  Score
0  Alice    85
1   Bob    92
      Name  Score
4   Eva    88
5  Frank    72
6  Grace    91

```

```

import pandas as pd
import matplotlib.pyplot as plt
data = {
    'Name': ['Arun', 'Bala', 'Chitra', 'Divya', 'Ezhil', 'Fathima'],
    'Department': ['CSE', 'ECE', 'CSE', 'ECE', 'CSE', 'ECE'],
    'Marks': [85, 78, 92, 88, 75, 95]
}
df = pd.DataFrame(data)
print("Original Data:")
print(df)
print("\nSorted by Marks:")
sorted_df = df.sort_values(by='Marks', ascending=False)
print(sorted_df)

```

```

Original Data:
      Name Department  Marks
0   Arun         CSE     85
1   Bala         ECE     78
2  Chitra         CSE     92
3  Divya         ECE     88
4  Ezhil         CSE     75
5 Fathima         ECE     95

Sorted by Marks:
      Name Department  Marks
5 Fathima         ECE     95
2  Chitra         CSE     92
3  Divya         ECE     88
0   Arun         CSE     85
1   Bala         ECE     78
4  Ezhil         CSE     75

```

```
import pandas as pd
import matplotlib.pyplot as plt
data = {
    'Name': ['Arun', 'Bala', 'Chitra', 'Divya', 'Ezhil', 'Fathima'],
    'Department': ['CSE', 'ECE', 'CSE', 'ECE', 'CSE', 'ECE'],
    'Marks': [85, 78, 92, 88, 75, 95]
}
df = pd.DataFrame(data)
print("Original Data:")
print(df)
print("\nAverage Marks by Department:")
grouped_df = df.groupby('Department')['Marks'].mean()
print(grouped_df)
```

Original Data:

	Name	Department	Marks
0	Arun	CSE	85
1	Bala	ECE	78
2	Chitra	CSE	92
3	Divya	ECE	88
4	Ezhil	CSE	75
5	Fathima	ECE	95

Average Marks by Department:

Department	Marks
CSE	84.0
ECE	87.0

Name: Marks, dtype: float64

```
import pandas as pd
import matplotlib.pyplot as plt
data = {
    'Name': ['Arun', 'Bala', 'Chitra', 'Divya', 'Ezhil', 'Fathima'],
    'Department': ['CSE', 'ECE', 'CSE', 'ECE', 'CSE', 'ECE'],
    'Marks': [85, 78, 92, 88, 75, 95]
}
df = pd.DataFrame(data)
print("Original Data:")
print(df)
df['Rank'] = df['Marks'].rank(ascending=False, method='dense')
print("\nStudents with Rank:")
print(df.sort_values(by='Rank'))
```

Original Data:

	Name	Department	Marks
0	Arun	CSE	85
1	Bala	ECE	78
2	Chitra	CSE	92
3	Divya	ECE	88
4	Ezhil	CSE	75
5	Fathima	ECE	95

Students with Rank:

	Name	Department	Marks	Rank
5	Fathima	ECE	95	1.0
2	Chitra	CSE	92	2.0
3	Divya	ECE	88	3.0
0	Arun	CSE	85	4.0
1	Bala	ECE	78	5.0
4	Ezhil	CSE	75	6.0

```
import pandas as pd
import matplotlib.pyplot as plt
data = {
    'Name': ['Arun', 'Bala', 'Chitra', 'Divya', 'Ezhil', 'Fathima'],
    'Department': ['CSE', 'ECE', 'CSE', 'ECE', 'CSE', 'ECE'],
    'Marks': [85, 78, 92, 88, 75, 95]
}
df = pd.DataFrame(data)
print("Original Data:")
print(df)
plt.figure(figsize=(8, 5))
plt.bar(df['Name'], df['Marks'])
plt.title('Student Marks')
plt.xlabel('Student Name')
plt.ylabel('Marks')
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
grouped_df = df.groupby('Department')['Marks'].mean()
plt.figure(figsize=(8, 5))
grouped_df.plot(kind='bar')
plt.title('Average Marks by Department')
plt.xlabel('Department')
plt.ylabel('Average Marks')
```

```
plt.xticks(rotation=0)
plt.tight_layout()
plt.show()
```

Original Data:

	Name	Department	Marks
0	Arun	CSE	85
1	Bala	ECE	78
2	Chitra	CSE	92
3	Divya	ECE	88
4	Ezhil	CSE	75
5	Fathima	ECE	95

